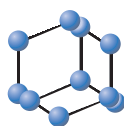


SYSTEMATIC-REVIEW ARTICLE

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SCIENCE

Efficiency of Glucosamine in Treating Temporomandibular Joint Osteoarthritis: A Meta-Analytic Umbrella Review



Sasidharan Sivakumar^{1,2}, Prabhakaran Prem Kumar³, Palindla Lakshmi Prasanna⁴, Gowardhan Sivakumar⁵, Sesuraj Balasamy⁶ and Ashok K. Sundramoorthy^{6,*}

¹Department of Public Health Dentistry, Saveetha Dental College and Hospitals, Saveetha Institute of Medical and Technical Sciences, 162, Poonamallee High Rd, Velappanchavadi, Chennai, Tamil Nadu, 600077, India; ²All India Institute of Medical Sciences, Sri Aurobindo Marg, Ansari Nagar, Ansari Nagar East, New Delhi, 110029, India; ³Department of Public Health Dentistry, Best Dental Science College, 69/1 Ultra Nagar, Madurai, Tamil Nadu, 625104, India; ⁴Department of Orthodontics & Dentofacial Orthopedics, Adhiparashakthi Dental College and Hospital, Melmaruvathur, Tamil Nadu, 603319, India; ⁵Private Practitioner, Sivam Dental & Maxillofacial Clinic, Guduvanchery, Tamil Nadu, 603202, India; ⁶Department of Prosthodontics, Saveetha Dental College and Hospitals, Saveetha Institute of Medical and Technical Sciences, 162, Poonamallee High Rd, Velappanchavadi, Chennai, Tamil Nadu, 600077, India

Abstract: Background: Temporomandibular joint osteoarthritis (TMJ OA) is a chronic disease that is a consequence of undue occlusal forces and is characterized by irreversible damage to the articular surfaces. Symptomatic slow-acting so-called nutraceutical drugs have been proposed as a treatment for osteoarthritis in comparison to non-steroidal anti-inflammatory drugs (NSAIDs). Oral glucosamine and chondroitin, slow-acting drugs, have been found to reduce pain and increase mouth opening in patients with TMJ OA. However, there is limited scientific evidence to confirm their clinical effectiveness.

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Aim: This systematic review was conducted to bolster the evidence supporting the assessment of the efficacy of glucosamine in the context of temporomandibular joint osteoarthritis (TMJ OA).

Methodology: This review identified four review articles from databases like Medline (via PubMed), Web of Science, Scopus, and EMBASE till September 2023 after screening at the title, abstract, and full-text level. They were assessed for risk of bias with the JBI risk of bias assessment tool.

Results: This review with meta-analysis focused on pooled estimate mean differences, revealing non-significant but discernible effects of glucosamine on maximum mouth opening (SMD = 0.288, $p = 0.15$) and pain reduction (SMD = 0.217, $p = 0.476$) in TMJ-related disorders.

Conclusion: Compared to control groups with ibuprofen and tramadol, glucosamine showed slightly more favourable outcomes. However, the variability in methodology and study characteristics warrants further longitudinal studies to confirm its efficacy.

Keywords: Osteoarthritis, glucosamine, systematic review, temporomandibular joint, splint therapies, chronic disease.

1. INTRODUCTION

Osteoarthritis (OA) is a chronic, prevalent disease that mostly affects joints that bear weight, such as the knees. When the dental specialty is concerned, the temporomandibular joint with increased loading was more prone to osteoarthritis for a few decades. Temporomandibular disorders are currently recognised as a major cause of non-dental pain in the orofacial region [1]. It is typified by a gradual

loss of cartilage and an inflammatory process that is followed by secondary alterations, such as the development of osteophytes and calcium deposition. Conventional treatment includes analgesics or non-steroidal anti-inflammatory drugs; however, lifestyle changes should also be recommended, such as weight reduction and specific exercises. Glucosamine and chondroitin, not classified as a drug, are classed as nutraceuticals, often self-administered by patients with OA [2]. Both agents are produced endogenously in the human body and are essential components of cartilage.

Since glucosamine was more frequently used in practice than chondroitin in many of the literature articles, our review prioritised glucosamine over chondroitin for assessment. Glucosamine stimulates the production of proteogly-

*Address correspondence to this author at the Department of Prosthodontics, Saveetha Dental College and Hospitals, Saveetha Institute of Medical and Technical Sciences, 162, Poonamallee High Rd, Velappanchavadi, Chennai, Tamil Nadu 600077, India; E-mails: ashokkumars.sdc@saveetha.com; ashok.sundramoorthy@gmail.com

cans by acting as a substrate for glycosaminoglycan synthesis [3]. Oral glucosamine is one of the symptomatic slow-acting drugs for osteoarthritis. Glucosamine may inhibit the production or activity of inflammatory molecules, such as cytokines (e.g., interleukin-1 β , tumour necrosis factor- α) and prostaglandins, thereby reducing inflammation and its detrimental effects on joint tissues [4]. Matrix metalloproteinases (MMPs) are enzymes that degrade extracellular matrix components, including collagen and proteoglycans, in joint tissues. Excessive MMP activity contributes to cartilage breakdown and inflammation in osteoarthritis. Glucosamine has been reported to inhibit MMP production or activity, thereby helping to preserve the structural integrity of cartilage and reduce inflammation within the joints. Alongside, it has been shown to possess antioxidant properties, scavenging ROS and reducing oxidative damage to joint tissues [5].

Based on the literature, there is not enough scientific evidence to unambiguously confirm the clinical effectiveness of glucosamine supplements in the treatment of temporomandibular joint osteoarthritis (TMJ OA). The most important aspect affecting oral glucosamine's clinical efficacy in treating TMJ OA was the total administration time. Administration of oral glucosamine for a longer period, *i.e.*, 3 months, led to a significant reduction in TMJ pain and a significant increase in maximum mouth opening [6]. It also resulted in long-term anti-inflammatory effects within the TMJs.

Research examining the effectiveness of glucosamine in temporomandibular joint osteoarthritis, especially in the form of systematic reviews (SRs), has grown considerably over the past decade. A network meta-analysis examining non-invasive pharmacological approaches for temporomandibular joint (TMJ) disorders highlighted the favourable outcomes associated with glucosamine-based management strategies. While the comparative efficacy did not reach statistical significance, glucosamine demonstrated potential as an effective intervention for TMJ disorders, contributing positively to prognosis and symptom alleviation [7]. However, the effectiveness of existing evidence remains inconclusive, and to our knowledge, no comprehensive literature has been conducted like any umbrella review. Hence, the objective of this study was to perform a systematic review of systematic reviews of randomized controlled trials along with quantitative data synthesis to evaluate the efficiency of glucosamine specifically in the context of temporomandibular joint osteoarthritis.

2. MATERIALS AND METHODS

2.1. Review Question

This systematic review was conducted for the following research question: “Does glucosamine have any influence in improvising the functional parameters of patients with temporomandibular joint osteoarthritis in comparison with any other drugs/ placebo?”. The study protocol can be accessed through the International Prospective Register of Systematic Reviews, the PROSPERO database, with the following register number: CRD42023472857.

2.2. Inclusion and Exclusion Criteria

The eligibility criteria were decided considering the following PICOS definition:

- **(P) Population:** Patients with osteoarthritis in Temporomandibular joint.
- **(I) Intervention:** Glucosamine was used for the treatment of osteoarthritis.
- **(C) Comparison:** Other drugs or placebo.
- **(O) Outcomes:** Increase in function.
- **(S) Study design:** Systematic reviews of Randomized controlled trials (RCT).

Original studies that addressed the “Subjects Intervention Control Outcome” parameters as per PICOS were eligible for inclusion. Systematic reviews of *ex-vivo* cell studies, case reports/series, and observational studies were all excluded.

2.2.1. Type of the Study

This review included reviews of human randomized experimental trials, comparing the effect of glucosamine with any other placebo or positive control in improving the functional parameters of the oral cavity.

2.3. Search Strategy for Article Identification

Two independent researchers conducted an electronic search for the systematic reviews *via* four databases, namely Medline (*via* PubMed), Web of Science, Scopus, and EMBASE, till September 2023. The search strategy for each database is summarised in Table 1. Only language-related limits were applied to the search, and only publications published in English were taken into consideration for this study.

Table 1. Search strategy for the study.

Database	Nos.	Search Strategy
Medline (PubMed)	4	((Temporomandibular joint[Title/Abstract]) OR (TMJ[Title/Abstract]) AND (Glucosamine[Title/Abstract])) AND (osteoarthritis[Title/Abstract])
Web Of Science (WoS)	4	((((TI="Temporomandibular joint" OR AB="Temporomandibular joint")) OR ((TI="TMJ" OR AB="TMJ")))) AND ((TI="Glucosamine" OR AB="Glucosamine")) AND ((TI="osteoarthritis" OR AB="osteoarthritis"))
Scopus	4	((TITLE-ABS("Temporomandibular joint")) OR (TITLE-ABS("TMJ"))) AND (TITLE-ABS("Glucosamine")) AND (TITLE-ABS("osteoarthritis"))
EBSCO	3	((('Temporomandibular joint':ti,ab) OR (TMJ:ti,ab)) AND (Glucosamine:ti,ab)) AND (osteoarthritis:ti,ab)

Table 2. General characteristics of the included studies.

Author, Year	Study Name	Study Setting	Participants (N)	Sources Searched	Test Group	Comparison Group
De Souza R.F, <i>et al.</i> , 2012 [26]	Cochrane review	Brazil	39	Cochrane Oral Health Group Trials Register, Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, EMBASE, PEDro	Glucosamine sulphate	Ibuprofen
Melo G, <i>et al.</i> , 2018 [13]	Systematic review	Brazil	159	EMBASE, LILACS, LIVIVO, PubMed, ScienceDirect, Cochrane Library, and Web of Science, Grey literature	Glucosamine sulphate	Placebo/ NSAIDs
Ruiz-Romero V, <i>et al.</i> , 2022 [12]	Systematic review and meta-analysis	Spain	139	Medline, Scopus, Cochrane library	Chondroitin sulphate and glucosamine medication	Other drugs (<i>i.e.</i> analgesics, <i>etc.</i>)
Derwich <i>et al.</i> , 2023 [6]	Systematic Review	USA	538	PubMed, Scopus	Glucosamine sulphate/ hydrochloride and Chondroitin sulphate	Placebo/ Analgesics

2.4. Screening of Articles

Upon pooling the articles from various databases, considering the similarities in title, author, and year of publication, the duplicates were removed. The titles and abstract level of screening were performed independently by two authors (S.S. and P.K.). Studies that failed to abide by the specified PICOS criteria were excluded following the author's agreement. Disagreements related to the selection of articles were resolved *via* discussion and consultation with a fourth author (G.S). Then, the full text of the eligible articles was screened individually by two authors (S.S. and P.K.), and the disagreement was resolved *via* discussion.

2.5. Data Extraction

For all studies that fulfilled the eligibility criteria, data extraction was performed by two authors (S.S and P.K) with the parameters in concordance with the Joanna Briggs Institute (JBI) methodological guide for umbrella review [8], which included: (a) author and year, (b) study name, (c) study setting, (d) participants, (e) sources searched, (f) test group, (g) comparison group, (h) range of years included, (i) number of studies included in a review, (j) appraisal instrument, (k) outcomes assessed, (l) results/ findings, (m) significance direction, (o) heterogeneity, (p) period of evaluation, and (q) inference, these are tabulated in Tables 2 and 3.

2.6. Risk of Bias

The studies were independently assessed for Risk of Bias (ROB) using the JBI systematic review critical appraisal tool [9] by two review authors, and disagreement among them was resolved by the fourth author (G.S).

2.7. Effect Measure and Statistical Analysis

Comprehensive Meta-Analysis software version 3.0, Biostat Inc. USA [10] was used for meta-analysis of selective studies which has outcomes comparable to those of systematically reviewed systematic review articles. In our analysis, we drew upon quantitative pooled estimate values from two of the four systematic review studies included in our research. These pooled estimates represent mean differences

in key clinical parameters, specifically maximum mouth opening and pain reduction. The random effect model was selected because of the expected heterogeneity between the included studies. Statistical heterogeneity was estimated using I^2 analyses, interpreting an I^2 value greater than 50% as a significant level of heterogeneity [11]. Statistical significance was determined as $p < 0.05$ for all analyses.

3. RESULTS

3.1. Study Selection and its Characteristics

The two authors (S.S. and P.K.) were independently involved in the study selection. After screening at the title and abstract level, followed by full-text screening based on the inclusion and exclusion criteria, four articles were finally processed for data extraction (Fig. 1). Based on the JBI methodological guide, the data extracted are tabulated in Tables 2 and 3. Table 3 includes the summary of outcome measures from the studies included in the review. The predominant studies evaluating TMJ-related complaints were in the functional parameters of maximum mouth opening limit and pain during its action. The predominant geographical locations where the systematic reviews were conducted are from the USA and Europe. The mean period of administration of glucosamine varies from 2.3 months to a maximum of 6.18 months.

3.2. Main Outcome of the Study

Among the 4 systematic reviews included in our review, only one study by Ruiz-Romero *et al.*, 2022 [12] performed a quantitative meta-analysis result with 3 RCTs included in it, which suggests a noteworthy upward trend in the improvisation of the mouth opening function. However, there was no effect on pain relief from glucosamine administration.

Out of all the review articles that were included, only one [6] that was published recently had the most articles ($n = 8$) to produce the evidence. The maximum mean follow-up period was 6.8 months, and this suggests that there was progress in both pain reduction and mouth-opening function.

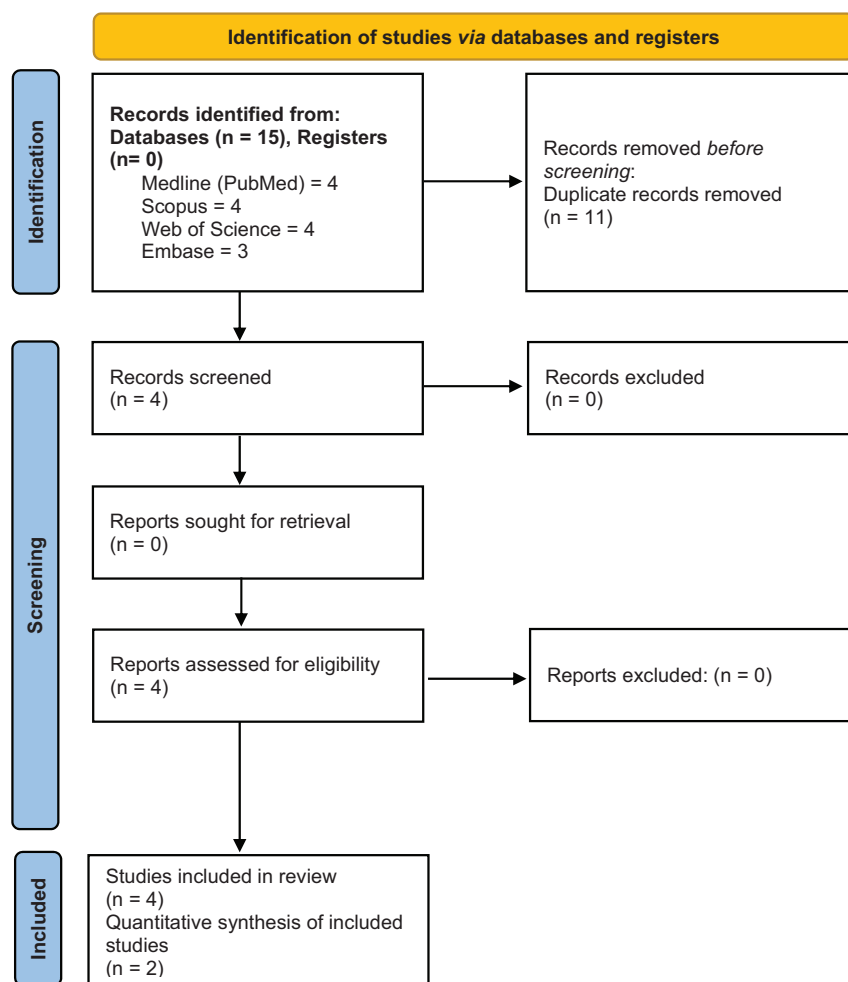


Fig. (1). The PRISMA flowchart.

In a study by Melo *et al.*, 2018 [13], it was mentioned that the lesser duration, like 6 weeks of glucosamine intake, was not superior to the placebo drug; instead, it was proved in the study by Derwich *et al.*, 2023 [6] that the duration of minimum 3 months was essential to observe a considerable difference in the improvement of TMJ related functional parameter.

A Cochrane review involving numerous pharmacological comparisons was released in 2012. Of them, only one randomised study included a glucosamine-based evaluation, which was likewise not significant in demonstrating the efficacy of glucosamine in treating TMJ osteoarthritis. It was after the year 2012 that several systematic reviews were undertaken on this topic of interest, which collectively support the usage of glucosamine as an adjunct to conventional measures in the management of TMJ disorders.

3.3. Risk of Bias

The JBI critical appraisal tool for systematic review was used to assess the risk of bias of the individual systematic reviews included in this review.

In summary, 11 criteria were considered for the assessment of ROB on the nominal scale of three sections: yes (present), no (absent), and not applicable. Among the 4 studies evaluated for ROB, predominantly all the evaluated criteria were present in all the included studies. The sole criterion that is not applicable to the included studies was the likelihood of measuring the publication bias, the reason being that a minimum of three studies would be needed and few papers would include just qualitative evidence synthesis. Thus, the risk of bias can be considered low with all the included studies (Table 4).

3.4. Quantitative Synthesis

Meta-analysis was performed with two of the included four studies. No further comparison was possible with all the studies due to the lack of quantitative outcome measures. The outcome measure used for analysis includes the pooled estimate as mean differences in the meta-analysis of the included studies. Quantitative results on maximum mouth opening revealed a pooled estimate standardized mean difference of 0.288 with a 95% confidence interval of -0.109 to

Table 3. Findings from the included studies.

Author, Year	Range of Years	No. of Studies	Appraisal Instrument	Outcome Assessed	Results Findings (Effect Measures)	Significance Direction	Heterogeneity	Period of Evaluation	Inference
De Souza <i>et al.</i> , 2012 [26]	1929- 2011	1	Cochrane risk of Bias tool	Pain, MMO	Mean difference MMO=1.75 (-4.10, 7.60), Pain reduction=-4.57 (-9.91, 0.77)	Pain: $P = 0.094$, MMO: $P = 0.56$	-	90 days	Glucosamine was as effective as ibuprofen in terms of pain, discomfort and jaw movements, according to one of the trials.
Melo <i>et al.</i> , 2018 [13]	2001- 2013	3	Cochrane risk of bias tool	Pain, MMO	Only descriptive, qualitative results were produced.	Neutral	-	10 weeks	Overall observation favours GS, but one of the studies suggested that GS was not superior to placebo, which can be attributable to the shorter administration period. Considering a study where the follow-up was 12 weeks, GS was found to be as effective as ibuprofen taken two or three times a day.
Ruiz-Romero <i>et al.</i> , 2022 [12]	2001- 2018	3	Cochrane risk of bias tool	Pain, MMO	Mean difference MMO=2.03 (-0.98, 5.04), Pain reduction=-0.07 (-0.71, 0.57)	For MMO - Positive, for pain reduction - Negative	I^2 MMO=86%, Pain reduction=0%	2.3 months	The combination of CS and GS seems to be effective in the functional improvement of TMD, and reduction of inflammatory markers, as well as an increase in MMO, without any adverse effects.
Derwich <i>et al.</i> , 2023 [6]	2001-2021	8	Cochrane risk of bias tool	Pain, MMO, levels of inflammatory markers	Only descriptive, qualitative results were produced. Significant pain reduction and increases the mouth opening	2 studies= Neutral, 6 studies= Positive	-	6.18 months	The factor affecting oral GS efficacy in treating TMJ OA was the administration time. Administration for a longer period of time, <i>i.e.</i> , 3 months, led to a significant reduction of TMJ pain and a significant increase in mouth opening. It also resulted in long-term anti-inflammatory effects within the TMJs.

Table 4. Risk of bias assessment of the included studies.

S. No.	Author, Year	De Souza <i>et al.</i> , 2012	Melo <i>et al.</i> , 2018	Ruiz-Romero <i>et al.</i> , 2022	Derwich <i>et al.</i> , 2023
	Type of Study	Cochrane Review with Meta-analysis	Systematic Review	Systematic Review and Meta-analysis	Systematic Review
1	Is the review question clearly and explicitly stated?	yes	yes	yes	yes
2	Were the inclusion criteria appropriate for the review question?	yes	yes	yes	yes
3	Was the search strategy appropriate?	yes	yes	yes	yes
4	Were the sources and resources used to search adequate?	yes	no	yes	yes
5	Were the criteria for appraising studies appropriate?	yes	yes	yes	yes
6	Was the critical appraisal conducted by two or more reviewers independently?	yes	yes	yes	yes
7	Were the methods to minimize errors in data extraction?	yes	yes	yes	yes
8	Were the methods used to combine studies appropriate?	yes	yes	yes	yes
9	Was the likelihood of publication bias assessed?	Not applicable	Not applicable	Not applicable	Not applicable
10	Were the recommendations for policy and/ or practice supported by the research data?	yes	yes	yes	yes
11	Were the specific directives for new research appropriate?	yes	yes	yes	yes

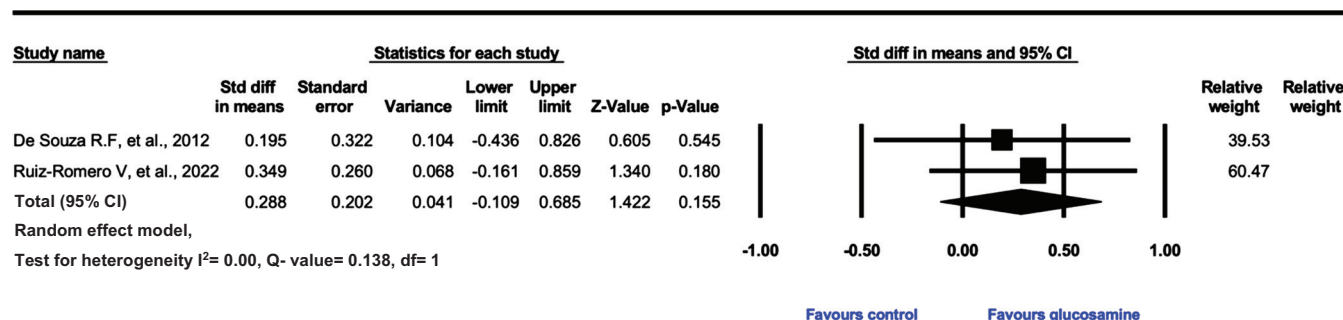


Fig. (2). Forest plot analysis of maximum mouth opening across interventions. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

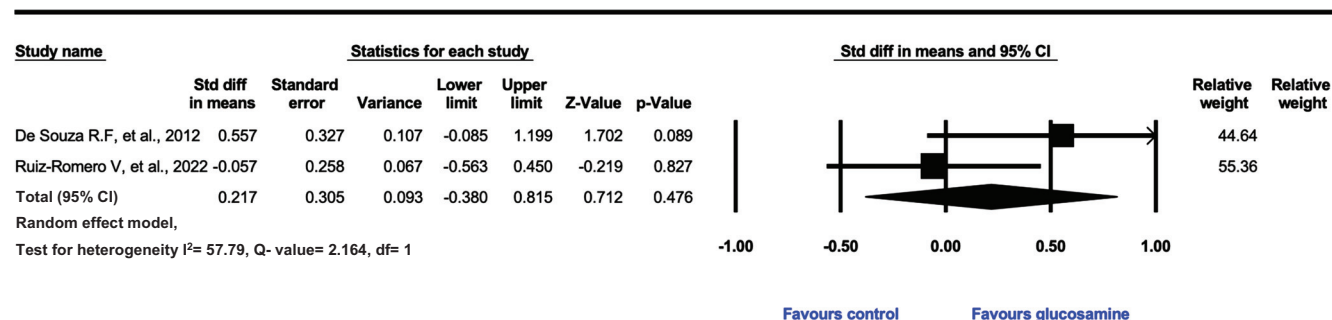


Fig. (3). Forest plot analysis of pain reduction across interventions. (A higher resolution / colour version of this figure is available in the electronic copy of the article).

0.685 and a significance value of 0.15 (Fig. 2). Similarly, quantitative results on reduction in pain score revealed a pooled estimate standardized mean difference of 0.217 with a 95% confidence interval of -0.380 to 0.815 and a significance value of 0.476 (Fig. 3).

These findings suggest that while the effect of glucosamine administration on maximum mouth opening and pain reduction may not reach statistical significance, there is still a discernible impact. This effect, though not overwhelmingly strong, indicates a trend toward improvement in the prognosis of TMJ-related disorders among subjects treated with glucosamine. On average, participants in the glucosamine group demonstrated slightly more favourable outcomes in both maximum mouth opening and pain reduction compared to those in the control group, which received treatment with ibuprofen and tramadol, as revealed in our analysis.

4. DISCUSSION

The present review considers the systematic reviews of randomized controlled trials that evaluated the influence of glucosamine in TMJ osteoarthritis. Glucosamine hydrochloride (GH) may decrease inflammation and pain by inhibiting

the synthesis of some joint-degenerating products, such as prostaglandin E2 and interleukin-1b, and thus may also prevent cartilage damage [14]. Chondroitin is a glycosaminoglycan and a major proteoglycan in joint cartilage. It can maintain viscosity in joints and stimulate cartilage repair and collagen synthesis by inhibiting lysosomal enzymes [15]. Although NSAIDs and some lifestyle changes were commonly advised for patients with TMJ osteoarthritis, glucosamine, and chondroitin, classed as over-the-counter supplements or nutraceuticals, are regularly self-administered by patients with OA. It is customary in a few other nations as well to prescribe these as supplemental measures to complement traditional medical care. Glucosamine hydrochloride is the first glucosamine product to be licensed as a medicine in the UK for symptomatic relief of mild to moderate osteoarthritis of the knee. Patients with osteoarthritis widely use glucosamine as a nutraceutical in the USA and as a pharmaceutical in Europe [16]. Table 2 depicts this in a clearer way: most of the systematic studies we included were conducted mostly in European and American nations where using glucosamine to treat osteoarthritis is more common.

There were numerous controlled trials evaluating glucosamine (oral, intramuscular, or intravenous administration) in comparison with placebo in patients with knee OA.

Still, those results cannot be replicated similarly for TMJ disorders because the articular surface of TMJ consists of dense fibrous connective tissue (fibrocartilage) when compared to the hyaline cartilaginous articular surfaces of other synovial joints [17]. The first clinical trial on evaluating the effects of glucosamine on TMJ disorders, published in the year 2001, revealed that in comparison with the ibuprofen, the GS group showed a significant improvement in pain and all the evaluated functional parameters [18]. Glucosamine was evaluated only in approximately seven clinical trials from 2001 to 2019 to assess its efficiency on osteoarthritis of TMJ, which provided inconclusive efficiency on TMJ osteoarthritis [4, 19-24]

Treatment efficiency with glucosamine was greatly affected by one parameter, the total administration period, which was even strengthened with evidence from two systematic reviews of our review by Melo *et al.*, 2018 [13] and Derwich *et al.*, 2023 [6], that a minimum of 3 months of administration is necessary for effective action of glucosamine in osteoarthritis when it is compared against the non-steroidal anti-inflammatory drugs (NSAIDs) or any other placebo drugs. As far as safety is concerned, there are a few unusual side effects of glucosamine, including [25, 26] heartburn, diarrhoea, nausea, and epigastric pain, and there are no documented interactions with other medications that exist for glucosamine. Thus, in clinical practice, glucosamine and chondroitin sulphate can be highly suggested for TMJ osteoarthritis.

4.1. Limitations

The limitations of this review include the fact that only four systematic reviews were included in the analysis, which may limit the generalizability of the findings. Additionally, the studies included in the systematic reviews varied in terms of their methodology, sample size, and study duration, which may have introduced heterogeneity in the results. Furthermore, the limited number of clinical trials specifically evaluating glucosamine for TMJ disorders highlights a gap in the existing literature, warranting caution in extrapolating findings to this specific patient population.

Despite the acknowledged limitations, the evidence synthesized from this review indicates that glucosamine can serve as a safe and effective adjunctive therapy in managing TMJ osteoarthritis (OA) with no reported adverse events. While initial clinical trials have demonstrated notable pain improvement, further research is imperative to ascertain its consistent efficacy, especially in the context of TMJ disorders, given the distinct fibrocartilaginous composition of TMJ articular surfaces. Consequently, glucosamine warrants inclusion in the comprehensive treatment regimen for TMJ OA patients, and clinicians should prioritize conducting longitudinal studies to rigorously evaluate its therapeutic efficacy.

CONCLUSION

The latest systematic review underscores the clinical applicability of glucosamine and chondroitin in managing tem-

poromandibular joint (TMJ) osteoarthritis (OA), with notable improvements in functional indicators, such as mouth opening and pain reduction. Importantly, their lower risk of side effects compared to traditional NSAIDs or COX-2 inhibitors, even with long-term use, highlights their potential as safer alternatives in clinical practice. When combined with minimally invasive therapies or conventional splints, these nutraceuticals offer a promising adjuvant approach for enhancing patient outcomes in TMJ OA management. However, investigations into potential mechanisms of action and patient-specific responses are essential for refining personalized treatment strategies. Overall, the evidence supports the consideration of glucosamine and chondroitin sulphate as valuable additions to the armamentarium for TMJ OA management, paving the way for improved patient care and outcomes in clinical practice.

AUTHORS' CONTRIBUTIONS

It is hereby acknowledged that all authors have accepted responsibility for the manuscript's content and consented to its submission. They have meticulously reviewed all results and unanimously approved the final version of the manuscript.

LIST OF ABBREVIATIONS

NSAIDs = Non-Steroidal Anti-Inflammatory Drugs

OA = Osteoarthritis

TMJ = Temporomandibular Joint

CONSENT FOR PUBLICATION

Not applicable.

STANDARDS OF REPORTING

PRISMA guidelines and methodology were followed.

AVAILABILITY OF DATA AND MATERIALS

The data and supportive information are available within the article.

FUNDING

None.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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Declared none.

SUPPLEMENTARY MATERIAL

PRISMA checklist is available as supplementary material on the publisher's website along with the published article.

REFERENCES

- [1] Guzman U, Gremillion HA. Temporomandibular disorders: Epidemiologic and etiologic considerations (Part 1). *Dental Tribune* 2007; 200(3): 8-10.
- [2] Huskisson EC. Glucosamine and chondroitin for osteoarthritis. *J Int Med Res* 2008; 36(6): 1161-79.
<http://dx.doi.org/10.1177/147323000803600602> PMID: 19094424
- [3] Cribbs B. Glucosamine: An alternative therapy for osteoarthritis?. *Nutrition Bytes* 1999; 5(2): 1-4.
- [4] Damlar İ, Esen E, Tatlı U. Effects of glucosamine-chondroitin combination on synovial fluid IL-1 β , IL-6, TNF- α and PGE2 levels in internal derangements of temporomandibular joint. *Med Oral Patol Oral Cir Bucal* 2015; 20(3): e278-83.
<http://dx.doi.org/10.4317/medoral.20242> PMID: 25662545
- [5] Sivakumar S, Sivakumar G, Sundramoorthy AK. Effects of glucosamine in the temporomandibular joint osteoarthritis: A review. *Curr Rheumatol Rev* 2024; 20(4): 373-8.
<http://dx.doi.org/10.2174/0115733971283009231225144322> PMID: 38213156
- [6] Derwich M, Górski B, Amm E, Pawłowska E. Oral glucosamine in the treatment of temporomandibular joint osteoarthritis: A systematic review. *Int J Mol Sci* 2023; 24(5): 4925.
<http://dx.doi.org/10.3390/ijms24054925> PMID: 36902359
- [7] Christidis N, Al-Moraissi EA, Barjandi G, *et al.* Pharmacological treatments of temporomandibular disorders: A systematic review including a network meta-analysis. *Drugs* 2024; 84(1): 59-81.
<http://dx.doi.org/10.1007/s40265-023-01971-9> PMID: 38103150
- [8] Edoardo A, Ritin S, Christina G, Cheryl H, Hanan K, Patraporn T. Methodology for JBI umbrella reviews. Australia: Joanna Briggs Institute Reviewers' Manual 2014; pp. 1-34.
- [9] Aromataris E, Fernandez R, Godfrey C, Holly C, Khalil H, Tung-punkom P. Umbrella Reviews. JBI Manual for Evidence Synthesis. JBI 2020.
- [10] Borenstein M, Hedges L, Higgins J, Rothstein H. Comprehensive Meta-Analysis Version 30. Englewood, NJ: Biostat 2022.
- [11] Higgins JPT, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Stat Med* 2002; 21(11): 1539-58.
<http://dx.doi.org/10.1002/sim.1186> PMID: 12111919
- [12] Ruiz-Romero V, Toledano-Serrabona J, Gay-Escoda C. Efficacy of the use of chondroitin sulphate and glucosamine for the treatment of temporomandibular joint dysfunction: A systematic review and meta-analysis. *Cranio* 2022; 2022: 1-10.
<http://dx.doi.org/10.1080/08869634.2022.2076796> PMID: 35588467
- [13] Melo G, Casett E, Stuginski-Barbosa J, *et al.* Effects of glucosamine supplements on painful temporomandibular joint osteoarthritis: A systematic review. *J Oral Rehabil* 2018; 45(5): 414-22.
<http://dx.doi.org/10.1111/joor.12616> PMID: 29446485
- [14] Gouze JN, Bordji K, Gulberti S, *et al.* Interleukin-1 β down-regulates the expression of glucuronosyltransferase I, a key enzyme priming glycosaminoglycan biosynthesis: Influence of glucosamine on interleukin-1 β -mediated effects in rat chondrocytes. *Arthritis Rheum* 2001; 44(2): 351-60.
[http://dx.doi.org/10.1002/1529-0131\(200102\)44:2<351::AID-ANR53>3.0.CO;2-M](http://dx.doi.org/10.1002/1529-0131(200102)44:2<351::AID-ANR53>3.0.CO;2-M) PMID: 11229466
- [15] Pavelka K, Gatterova J, Olejarova M, Machacek S, Giacovelli G, Rovati LC. Glucosamine sulfate use and delay of progression of knee osteoarthritis: A 3- year, randomized, placebo-controlled, double-blind study. *Arch Intern Med* 2002; 162(18): e2123.
- [16] Shikhman AR. Glucosamine and osteoarthritis. *Int J Clin Rheumatol* 2006; 1(1): 67.
- [17] de Bont LGM, Liem RSB, Havinga P, Boering G. Fibrous component of the temporomandibular joint disk. *Cranio* 1985; 3(4): 368-73.
<http://dx.doi.org/10.1080/08869634.1985.11678119> PMID: 3864873
- [18] Thie NM, Prasad NG, Major PW. Evaluation of glucosamine sulfate compared to ibuprofen for the treatment of temporomandibular joint osteoarthritis: A randomized double blind controlled 3 month clinical trial. *J Rheumatol* 2001; 28(6): 1347-55.
PMID: 11409130
- [19] Cömert Kılıç S. Does glucosamine, chondroitin sulfate, and methylsulfonylmethane supplementation improve the outcome of temporomandibular joint osteoarthritis management with arthrocentesis plus intraarticular hyaluronic acid injection. A randomized clinical trial. *J Craniomaxillofac Surg* 2021; 49(8): 711-8.
<http://dx.doi.org/10.1016/j.jcms.2021.02.012> PMID: 33685850
- [20] Yang W, Liu W, Miao C, Sun H, Li L, Li C. Oral glucosamine hydrochloride combined with hyaluronate sodium intra-articular injection for temporomandibular joint osteoarthritis: A double-blind randomized controlled trial. *J Oral Maxillofac Surg* 2018; 76(10): 2066-73.
<http://dx.doi.org/10.1016/j.joms.2018.04.031> PMID: 29864430
- [21] Cen X, Liu Y, Wang S, Yang X, Shi Z, Liang X. Glucosamine oral administration as an adjunct to hyaluronic acid injection in treating temporomandibular joint osteoarthritis. *Oral Dis* 2018; 24(3): 404-11.
<http://dx.doi.org/10.1111/odi.12760> PMID: 28862770
- [22] Cahlin BJ, Dahlström L. No effect of glucosamine sulfate on osteoarthritis in the temporomandibular joints—a randomized, controlled, short-term study. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 2011; 112(6): 760-6.
<http://dx.doi.org/10.1016/j.tripleo.2011.06.012> PMID: 22001199
- [23] Nguyen P, Mohamed SE, Gardiner D, Salinas T. A randomized double-blind clinical trial of the effect of chondroitin sulfate and glucosamine hydrochloride on temporomandibular joint disorders: A pilot study. *Cranio* 2001; 19(2): 130-9.
<http://dx.doi.org/10.1080/08869634.2001.11746162> PMID: 11842864
- [24] Behnia A, Haghighat A, Kaviani N, Khorami B. Evaluation of Glucosamine sulfate and Ibuprofen effects in patients with temporomandibular joint osteoarthritis symptom. *J Res Pharm Pract* 2013; 2(1): 34-9.
<http://dx.doi.org/10.4103/2279-042X.114087> PMID: 24991602
- [25] Dahmer S, Schiller RM. Glucosamine. *Am Fam Physician* 2008; 78(4): 471-6.
PMID: 18756654
- [26] de Souza RF, da Silva CH, Nasser M, Fedorowicz Z, Al-Muharraqi MA. Interventions for managing temporomandibular joint osteoarthritis. *Cochrane Database Syst Rev* 2012; 2012(4): CD007261.
<http://dx.doi.org/10.1002/14651858.CD007261.pub2>